

Key-Driven Identity Protection and Secure Traceability for Face-Swapping

Miaomiao Wang , Sheng Li , Xinpeng Zhang , and Guorui Feng 

Abstract—Face swapping replaces the identity of a target face with that of a source face while retaining non-identity attributes, and is widely used in privacy protection scenarios. However, removing identity information makes it difficult to support subsequent recognition tasks. Although active tracing methods can determine content ownership, techniques such as watermarking rely on external data binding, which cannot establish identity associations in the feature space nor support authorization-controlled identity verification. To address this issue, we propose a key-driven Identity Protection and Secure Traceability (IPST) framework for face-swapped images, which enables identity binding and controlled verification without exposing visual identity. The proposed method operates as a plug-in applied to face-swapping results, introducing an implicit identity association in the feature space so that the target identity participates in the generation process while remaining visually imperceptible. Specifically, IPST consists of three stages. First, an identity transfer stage aligns the generated image with the target identity in the recognition space via feature-guided perturbations. Second, an identity hiding stage performs key-controlled latent space editing to map the identity into a concealed state, ensuring that the published image cannot be directly recognized. Finally, an identity recovery stage reconstructs the target identity under authorized conditions through inverse latent manipulation, enabling identity verification using face recognition models. Experimental results demonstrate that the proposed method achieves stable identity binding while preserving visual quality, and shows strong robustness against unauthorized recovery. Moreover, the framework is model-agnostic and can be seamlessly integrated with existing face swapping systems, making it suitable for practical deployment.

Index Terms—Latent space operation, identity preservation, identity tracking, active protection.

I. INTRODUCTION

WITH the rapid development of generative models, face-swapping has achieved significant progress in visual realism and semantic consistency [1]. Modern approaches can preserve pose, illumination, and expression while performing identity transfer, enabling applications in film production, digital content creation, and privacy-preserving face generation [2], [3]. However, along with these benefits, face-swapping also

introduces privacy and security risks, including identity misuse, misinformation, and portrait rights violations [4], [5].

To address these risks, existing studies mainly focus on passive detection and active defense. Passive detection methods [6], [7], [8], [9], [10], [11] aim to identify manipulated content by analyzing visual artifacts, which is useful for post-event analysis but cannot prevent misuse during content generation. Meanwhile, methods [12], [13] point out that counterfeiters can easily avoid post-event evidence.

Active defense methods introduce protection mechanisms before content generation or release and are generally categorized into active interference and active tracing. Active interference methods [14], [15], [16], [17], [18] attempt to disrupt face-swapping models by adding adversarial noise, which may degrade image quality or limit legitimate usage. The active tracking methods [19], [20], [21], [22], [23] protect the samples by embedding erasable and robust watermarks for detection and tracking, respectively. During the watermark verification process, the presence of erasable watermarks is used to determine whether forgery has occurred and sustainable watermarks are used for tracking. However, the cost of embedding watermarks into massive images on social media makes it difficult to promote.

In practical applications, face-swapping techniques have also been widely used for facial privacy protection to conceal real identities [24], [25], [26]. In this setting, the target face, as the protected subject, determines the semantic consistency and behavioral expression of the swapped image. During data sharing and content publication, the identity information of the target face remains imperceptible to meet privacy requirements. However, these methods are typically irreversible: the original identity is completely hidden after replacement and cannot be recovered, which limits their applicability in scenarios requiring identity verification or accountability. In this context, the target face is both a core input to the generation process and sensitive information that must be strictly protected. How to achieve verifiable and controllable identity association with the real individual without exposing identity has thus become a new research problem.

Under this setting, existing methods are insufficient. Active perturbation methods interfere with the generation process and are not suitable for privacy-preserving applications. Watermark-based tracing methods embed external identifiers rather than identity-related representations, and therefore cannot support identity-level verification. In addition, some approaches encode information related to facial features, which may lead to privacy

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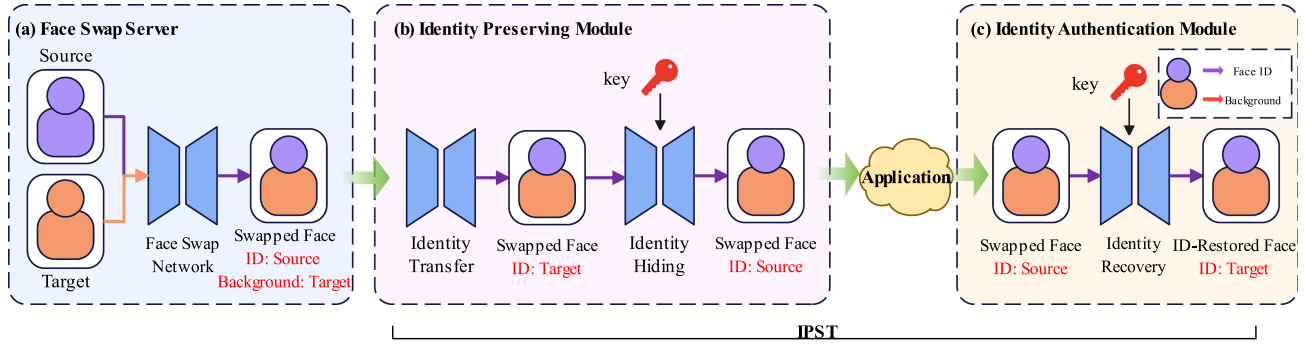


Fig. 1. Illustration on the application of our proposed IPST framework. The proposed legal and safe face-swapping network is mainly used to solve the problem of Deepfake infringement. When a suspected face-swapping dispute occurs, the target ID can be extracted through the key for identity authentication.

leakage during extraction. Furthermore, watermark extraction typically does not enforce access control, making it difficult to support authorized-only tracing. In some cases, the verification process also depends on third-party auditing. As a result, existing methods cannot satisfy identity participation, identity hiding and controllable traceability those scenarios. We consider a different problem from conventional Deepfake defense: enabling identity binding and controlled accountability at the image level under privacy constraints.

To this end, we propose a key-driven Identity Protection and Secure Traceability (IPST) framework. The proposed method is designed as a plug-in module that can be integrated with existing face-swapping systems. It establishes an implicit association between the generated image and the target identity without altering visual appearance, and uses a key mechanism to control access to identity information. As a result, identity information remains hidden during normal usage and can only be recovered and verified by authorized parties. Fig. 1 illustrates the IPST framework. Specifically, (1) a source face providing identity features and a target face providing non-identity attributes are processed by a face-swapping service to generate an image that preserves the source identity while retaining the target's background, pose, and expression; (2) the proposed identity protection module processes the swapped image before publication on social media, enabling identity tracing in case of malicious tampering; (3) for disputed content, the received image can be input into the identity verification module to perform identity protection and secure tracing.

The main contributions can be summarized as:

- We define the problem of identity binding and controlled traceability under privacy constraints in face-swapping applications, and propose a corresponding solution.
- We propose a key-driven Identity Protection and Secure Traceability (IPST) framework that enables implicit identity embedding, identity hiding, and controlled recovery at the image level.
- We provide a unified solution that supports identity traceability without exposing visual identity information, enabling authorized-only access.
- The proposed method can be integrated with different face-swapping models as a plug-in, supporting practical deployment in privacy-sensitive applications.

II. RELATED WORK

A. Defense for Active Interference

The defense technology based on active interference destroys the output of Deepfake models by injecting adversarial perturbations into the samples, thereby preventing the occurrence of Deepfake. From the perspective of technical means, it can be divided into three types of defense. (a) Methods based on data poisoning. It disrupts facial detectors based on deep neural networks by adding specially designed invisible perturbations to the original image [27]. This method is mainly applied in the training stage of Deepfake models, increasing the false detection rate and rejection rate by adding poisoned data to the training set, thereby reducing the availability of facial data and preventing the occurrence of face-swapping. Landmark Breaker [28] focuses on attacking facial feature point extractors, altering the predictions of facial feature points, thereby disrupting facial alignment and reducing the data quality of facial key-point extraction. This method causes the forged image to have obvious signs of tampering and lose its usability. (b) Methods based on adversarial attacks. The significant difference between this method and the method of data poisoning is that its defense mainly occurs in the application stage, when attackers attempt to forge protected images, the output is distorted. Anti-Forgery [17] explores how to use perceptual imperceptible disturbances to interfere with fake models, causing visual biases in the generated images and thereby reducing their authenticity. Guan et al. [29] improve the reliability of defense by increasing the difficulty of attacks. They study how to integrate multiple adversarial perturbations to prevent attackers from tampering with protected images. Yeh et al. [30] and Segalis et al. [31] discuss how to conduct adversarial attacks against image translation-based Deepfake algorithms. Those methods cannot only attack known Deepfake models but also resist the training process of the model. Even if the total number of samples is included in the training set, it can effectively destroy the results of Deepfake. (c) Methods based on latent space editing. The latent space-based methods [16], [32] do not directly add perturbations to the original face image. Firstly, the face samples that need to be protected are mapped to the hidden space, and then the image is edited specifically in the hidden space so that the forger cannot obtain the face hidden space inverse mapping that can be used for forgery, thereby

avoiding the risk of Deepfake. However, this method overly relies on Gan's inversion, which limits the range of adversarial perturbations by utilizing perturbation thresholds, but still leads to a decrease in facial visual quality.

B. Defense for Active Tracking

Unlike active interference technology, defense methods based on active tracking technology do not require the performance of Deepfake models to be compromised. On the contrary, they attempt to track the source of forged images or detect images that are difficult to determine their authenticity through watermark. Sepmark [19] and FakeTagger [20] track content sources and detect Deepfake by embedding source data and watermarks. FakeTagger [20] introduces a source tracking-based defense method for Deepfake propagation paths that can trace the propagation path of the original image even after the content is tampered with or forged. However, for unknown deep-forgery models, further training is required. Sepmark [19] is proposed as a depth separable watermark, which embeds imperceptible watermarks without sacrificing visual quality and is used for unified source tracking and detection. Zhao et al. [23] can verify and track the true source of content even if the image content is tampered with by embedding a unique watermark in the original media. This method provides an identity verification layer for media content, which can be traced back to the original source even after deeply forged images have been propagated multiple times. FakeTacer [22] adopts a unique training data pollution strategy to detect Deepfake. Introducing forged samples during the training process enables the detection model to better identify deeply forged content. SIDT [21] focuses on protecting the security of the source facial identity in Deepfake by implicitly embedding the source facial image in the forgery image, ensuring that the generated encoded image cannot be visually distinguished from the original forgery image. Even after sharing on social media, the embedded source face and its identity can still be restored through a decoder, providing an active protection mechanism for the source face identity.

C. Face-Swapping for Privacy Preservation

Face-swapping have recently been introduced into face privacy preservation scenarios. In these methods, the target face is treated as the protected subject. During the swapping process, identity-irrelevant attributes such as pose, expression, and background are preserved, while identity-related information is replaced by a source face or a randomly generated identity, enabling de-identification or identity anonymization.

Sun et al. [3] propose a hybrid model that combines traditional image processing with deep generative models to achieve stable identity replacement while preserving visual consistency. Kuang et al. [24] further focus on the imperceptibility of anonymization results, aiming to reduce identity leakage while maintaining visual naturalness. With the development of generative models, StyleGAN-based methods have become widely used.

StyleIPSB [25] disentangles identity and attribute representations in the semantic latent space, enabling controllable face-swapping with high fidelity. In addition, Bensaid et al. [26] propose a latent space layer swapping method that exchanges features across different layers of the generative model to achieve face anonymization while preserving facial attributes.

However, these methods typically achieve privacy protection by fully removing or replacing identity information, which breaks the linkage between the generated result and the original identity. As a result, in scenarios that require both privacy protection and identity verification or accountability, such as cross-platform authentication or face-based access control, these methods do not support subsequent identity association and verification.

D. Comparison With Existing Methods

As shown in Table I, existing active interference methods focus on preventing face-swapping generation. They disrupt the performance of face-swapping models by interfering with feature extraction via data poisoning [28] or injecting adversarial perturbations [17], resulting in distorted swapped images that fail to support legitimate face-swapping scenarios. Furthermore, these methods do not incorporate privacy protection or identity tracing mechanisms.

Existing active tracking methods are categorized into watermark [19], [20] and source face [21] embedding-based approaches. These methods can support content attribution to a certain extent. However, they fundamentally rely on binding external data to images, establishing a linkage between the content and an auxiliary identifier through additional information, rather than associating the image with identity in the facial feature space. As a result, they are not suitable for controlled identity-level verification based on face recognition models.

In contrast, we consider a different problem setting by focusing on the privacy of the target identity in swapped facial images during their release and usage, rather than protecting the face-swapping model. Specifically, we establish a controllable contact between privacy-preserving swapped facial images and the target identity. The identity remains undetectable during content publishing without affecting visual appearance, and can be recovered and recognized under key-authorized conditions.

III. PROBLEM DESCRIPTION

IPST is an extended framework for privacy-preserving face-swapping. It aims to protect identity information and enable controllable traceability without disrupting existing face-swapping pipelines. The entities participating in the system include legitimate users, the system and adversaries. Their actions are as follows:

- *User* intends to utilize legitimate face-swapping tools for identity protection, while enabling controllable identity verification when necessary.
- *System* provides face-swapping services and incorporates identity protection and controllable traceability mechanisms during the generation process.

TABLE I
COMPARISON BETWEEN IPST AND EXISTING ACTIVE DEFENSE METHODS

Dimension	Active Interference Methods [28], [17]	Active Tracking Methods [19], [20], [21]	Proposed IPST Model
Objective	Disrupt face-swapping to prevent fakes	Embed source face for traceability	Allow legal face-swapping, protect identity and authorize traceability
Intervention Stage	Training (data poisoning [28]) / Application (adversarial perturbations [17])	After swapping faces, before publishing	After swapping faces, before publishing
Identity Protection and Traceability	No identity protection and traceability	SIDT [21] leaks visual privacy, watermarking [19], [20] no active protection	No visual leakage, key-controlled traceability
Identity Protection and Traceability	None	None	Key control mechanism
Closed-Box Applicability	Partially applicable	Partially applicable	As a plugin, supporting closed-box
Large-Scale Cost	High (retraining [28]) / Moderate (perturbations [17])	High (computational cost for embedding)	Low (one-time key editing)

- **Adversary:** Unlike attackers who use face-swapping tools (especially open-source or illegal tools) to generate malicious Deepfake images (such as forging someone else's portrait), adversaries can obtain the published swapped facial images and attempt to recover, infer or abuse the target identity information.

A. Problem Statement

In privacy-preserving face-swapping scenarios, the target face possesses dual attributes: it determines the semantic appearance of the generated image (such as pose, expression, and background) and contains sensitive identity information that needs protection. Therefore, the target identity should remain imperceptible during content distribution, while supporting authentication and accountability when necessary.

Existing methods struggle to meet these requirements. On one hand, face-swapping methods are typically irreversible, making it impossible to recover the target identity. On the other hand, methods such as watermarking, hashing, and digital signatures rely on external information binding, focusing on copyright ownership rather than identity authentication. The copyright extraction process relies on the watermark extraction module lacking a key-controlled security mechanism.

To achieve this goal, the system needs to possess the following characteristics: The target identity is intrinsically linked to the swapped facial image in the form of features. The target identity is visually imperceptible and cannot be recovered without authorization. Target identity recovery is key-controlled, and only authorized parties can complete authentication and tracing.

B. Threat Model

We hypothesize that adversaries can obtain privacy-preserving swapped facial images and attempt to recover, infer, or misuse target identity information from them. Simultaneously, adversaries may perform authentication processes without authorization or bypass system mechanisms by tampering with input.

We focus on the following three main threats:

- **Unauthorized Identity Recovery:** Adversaries attempt to recover target identity information from generated images without possessing the legitimate key.

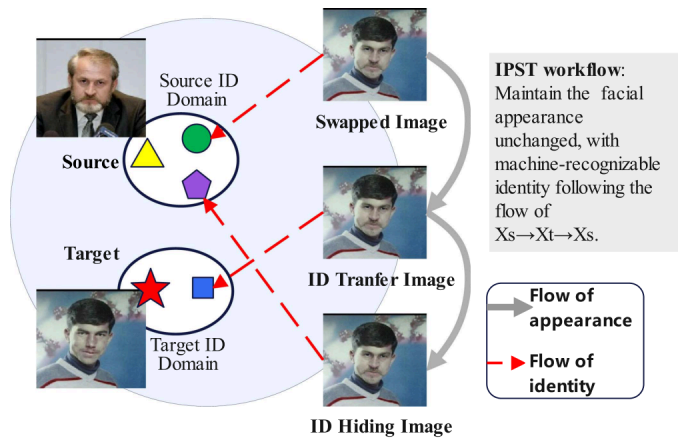


Fig. 2. The preserving process of target identity. It describes the IPST workflow during identity transfer and identity hiding. The appearance always remains consistent with the swapped facial image. The machine-recognizable identity first approaches X_s , then shifts to approach X_t , and finally returns to approach X_s .

- **Privacy Leakage Risk:** During identity embedding and recovery, identifiable information of the target's face may be accidentally exposed.
- **Authentication Abuse and Bypass:** Adversaries attempt to obtain valid authentication results using illegitimate input or incorrect keys.

To address these threats, the system must satisfy the following security requirements: under unauthorized conditions, the target identity should remain unrecoverable, the identity recovery process should be strictly controlled by keys, and the authentication results should be robust to illegitimate input, ensuring that only authorized parties can complete valid authentication.

IV. METHODOLOGY

A. Overview

We provide an overview of the proposed IPST framework. Based on existing face-swapping models, this framework introduces an identity processing module after the generation process to achieve mechanisms for identity binding, concealment, and controllable recovery. Fig. 2 shows the identity domain of the swapped facial image at each stage.

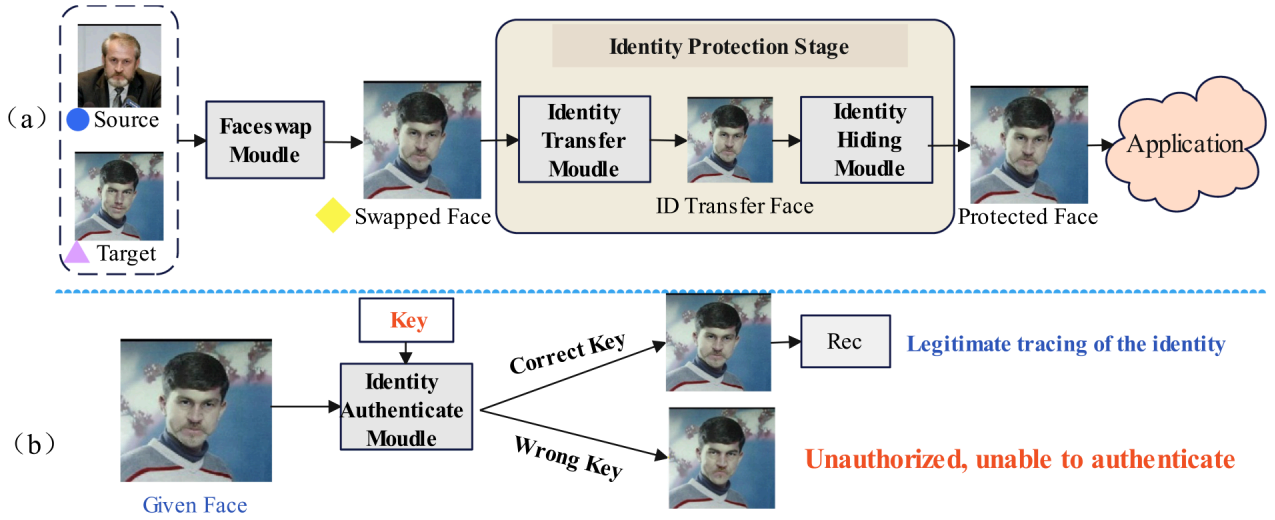


Fig. 3. The proposed overall framework for IPST. (a) Identity protection process: Add targeted adversarial perturbation and key-driven latent space editing to the swapped facial image for identity-preserving. (b) Identity tracking process: When using the correct key for identity tracking, the recovered facial image has the same appearance as the swapped facial image, and the identity belongs to the target identity. When using the wrong key for recovery, the recovered image will be distorted and cannot be used for identity tracking.

Unlike traditional methods that embed additional information (such as watermarks) into the image, IPST establishes the association between the target identity and the swapped facial image in the face feature space, making the target identity controllably recognizable by machines but visually imperceptible.

Specifically, given a source face X_s and a target face X_t , the face-swapping model first generates a swapped facial image X_F , whose identity-independent attributes are inherited from X_t , while its identity features inherit from X_s . Based on this, IPST further processes X_F to achieve the introduction and protection of the target identity.

Identity Transfer Module: We first apply a slight perturbation to the swapped image X_F to obtain $X_{F_{adv}}$, which gradually brings the image closer to the target identity X_t in the face recognition feature space, while maintaining a high degree of visual consistency with the original swapped image. At this point, the identity attribution can be described as follows:

$$I(X_{F_{adv}}) = I(X_t), \quad (1)$$

where I represents the facial recognition model.

After this stage of processing, the image has established a preliminary association with the target identity at the feature level, but this association can be directly perceived by the recognition model.

Identity Hiding Module: To prevent unauthorized access to the target identity, we further transform $X_{F_{adv}}$ in the latent space to obtain the final published image $X_{F_{k_1}}$.

This process, controlled by the user key k_1 , performs lightweight editing of the latent variables, remapping the identifiable identity back to the source identity while maintaining visual appearance unchanged. Therefore, to external observers or standard face recognition models, the published image still appears to represent the source identity, thus hiding the target

identity. It can be represented as:

$$I(X_{F_{k_1}}) = I(X_F) = I(X_s). \quad (2)$$

Identity Authenticate Module: In cases of content disputes or when identity verification is required, the system can restore the target identity under authorized conditions. Specifically, given a published image $X_{F_{k_1}}$ and a corresponding key k_1 , the identity authentication module reverses the latent space editing process to generate a restored image $X_{reF_{k_1}}$.

When the correct key is entered, the facial features are realigned to the target identity X_t , achieving identity verification based on a face recognition model. It can be represented as:

$$I(X_{reF_{k_1}}) = I(X_t). \quad (3)$$

Conversely, if an incorrect key k_2 is entered, the latent space restoration process will produce incorrect results, leading to significant image distortion or identity matching failure, thus preventing unauthorized identity inference. The formula is:

$$I(X_{reF_{k_2}}) \neq I(X_t). \quad (4)$$

B. The Proposed Framework

The overall structure of our proposed IPST framework is shown in Fig. 3. It is applied in identity protection and authentication processes, integrating a face-swapping network, an identity transfer module, an identity hiding module and an identity authentication module.

Face-Swapping Network: We use the pre-trained face-swapping network as server $\mathcal{F}$ to replace core area of the target face X_t with the source face X_s , resulting in a swapped facial image X_F , of which identity is similar to X_s in both visual perception and machine recognition. The process of face-swapping can be described as $X_F = \mathcal{F}(X_s, X_t)$.

Identity Transfer Module: We propose an identity transfer module constructed based on Generative Adversarial Networks

(GAN), consisting of a generator $\mathcal{G}_1$ and a discriminator $\mathcal{D}_1$. The core goal of this module is to add adversarial perturbations to the swapped facial image X_F , enabling the perturbed image to inherit the facial representation of the target face X_t while maintaining visual consistency with X_F . To avoid unnecessary pixel disturbances in non-critical regions, the generator $\mathcal{G}_1$ takes both the swapped image X_F and the source face X_s as inputs, with X_t serving as the identity condition. It generates a pixel-level weak adversarial perturbation mask $\mathcal{G}_1(X_F)$ that is specifically targeted at facial salient regions. The adversarial face image is formulated as: $X_{F_{adv}} = X_F + \mathcal{G}_1(X_F)$. This adversarial image $X_{F_{adv}}$ retains the same visual appearance as the original swapped image X_F , but possesses the same identity as the source face X_s , i.e., $I(X_{F_{adv}}) = I(X_t)$ (where $I(\cdot)$ denotes the identity output of a face recognition model).

Identity Hiding Module: We propose an identity hiding module constructed based on StyleGAN2 consisting of a pre-trained encoder $\mathcal{E}$, a latent space operator $\mathcal{O}$ and a pre-trained generator $\mathcal{G}_2$. The core goal of this module is to hide the identity of the target face X_t (inherited by the adversarial image $X_{F_{adv}}$) under the control of a user key k ensuring the final image maintains visual consistency with the swapped image X_F while its identity aligns with the source face X_s . First the adversarial image $X_{F_{adv}}$ (which inherits X_t 's identity) is mapped to the latent space via the encoder $\mathcal{E}$ to obtain a latent vector z the latent space operator $\mathcal{O}$ takes the user key k as a control parameter to fine-tune z (satisfying $\|z - z'\| < \epsilon$ to guarantee visual consistency) outputting the adjusted latent vector z' . Finally, the generator $\mathcal{G}_2$ performs an inverse mapping on z' to generate the identity-hidden image X_{F_k} . This module is cascaded with the identity transfer module through the adversarial image $X_{F_{adv}}$ and the key k is injected into the latent space editor as an independent parameter to achieve key-controlled hiding. Mathematically, the identity of X_{F_k} satisfies $I(X_{F_k}) = I(X_s)$ (where $I(\cdot)$ denotes the identity output of a face recognition model).

Identity Authentication Module: The identity authentication module shares the same model architecture as the identity hiding module. The core difference lies in that the operational logic of the latent space operator $\mathcal{O}$ is an inverse process. The image to be verified X_{F_k} and the user key k are input, and the latent vector z_k is obtained through the encoder $\mathcal{E}$. The latent space operator $\mathcal{O}$ restores the original latent vector z based on the correct key k_1 , and then the generator $\mathcal{G}_2$ generates the identity-restored image $X_{reF_{k1}}$. This image has the same identity as the target face X_t (i.e., $I(X_{reF_{k1}}) = I(X_t)$) and maintains consistency with the appearance of X_F . If an incorrect key k_2 is input, the editor $\mathcal{O}$ outputs an incorrect latent vector, leading to visual distortion of the restored image and mismatched identity (i.e., $I(X_{reF_{k2}}) \neq I(X_t)$), where $I(\cdot)$ denotes the identity output of a face recognition model.

C. Loss Functions

We design multiple optimization objectives aimed at satisfying the problems to be achieved in Section III. We use cosine embedding loss $\mathcal{L}_{\cos}$ to measure the similarity between two

facial features f_1 and f_2 , defined as follows:

$$L_{\cos}(f_1, f_2, l) = \begin{cases} (1 - \cos(f_1, f_2)) & , l = 1, \\ \max(m, \cos(f_1, f_2)) & , l = -1, \end{cases} \quad (5)$$

where m represents a hyperparameter, $l = 1$ means that f_1 and f_2 come from faces with the same identity, $l = -1$ means that f_1 and f_2 are from different faces.

1) Identity Transfer Module: Our goal is to add weak perturbations to the swapped facial image, so that the image with perturbations remains unchanged in appearance, but the identity is transferred and belongs to the same identity as the source face. It can be achieved through the following losses.

Perturbation Loss: To minimize the impact of added adversarial perturbations on image quality, we introduce a perturbation loss to constrain the size of the added perturbations. It can be described as:

$$\mathcal{L}_{per} = \max(\sigma, \|\mathcal{G}_1(X_F)\|), \quad (6)$$

where $\sigma \in [0, +\infty)$ is a hyperparameter that restricts perturbations, $\mathcal{G}_1(X_F)$ represents the added adversarial mask.

Identity Consistency Loss: In the phase of transferring the identity, we generate adversarial sample with the target identity by targeted adversarial attacks. To make the swapped facial image consistent with the target facial identity, we need a face recognition model $\mathcal{R}$ to supervise the training. We use the cosine similarity loss as the optimization objective to give higher similarity scores for two different identities. Its expression is:

$$\mathcal{L}_{id} = \cos(\mathcal{R}(X_{F_{adv}}), \mathcal{R}(X_t), 1). \quad (7)$$

Adversarial Loss: Since our goal is to obtain an adversarial example image that is visually close to the input image, we use adversarial training in this framework and introduce adversarial loss to constrain the perceptual authenticity of the generated image. Therefore, we use adversarial losses to distinguish between $X_{F_{adv}}$ and X_F

$$\mathcal{L}_{GAN} = \log \mathcal{D}(X_F) + \log(1 - \mathcal{D}(X_{F_{adv}})). \quad (8)$$

Total Loss: The total loss of this module is as follows:

$$\mathcal{L}_{total1} = \lambda_{per}\mathcal{L}_{per} + \lambda_{id}\mathcal{L}_{id} + \mathcal{L}_{GAN}, \quad (9)$$

where λ_{per} and λ_{id} are hyperparameters controlling the identity loss.

2) Identity Hiding Module: A key-driven latent space editing method is proposed to hide the source identity and maintain the appearance of the reconstructed image. Firstly, we calculate the potential representation of the adversarial swapped facial image $X_{F_{adv}}$ to obtain z . Then, the k and z are combined through operator $\mathcal{O}$ to obtain z' . When z and z' are similar, the reconstructed image X_{F_k} should be visually similar to the swapped facial image X_F . Therefore, it can be refined into:

$$\begin{aligned} \min_{z'} \mathcal{L}(X_F, X_{F_k}), \\ s.t. \quad l(z, z') < \epsilon, \end{aligned} \quad (10)$$

where X_{F_k} represents the reconstructed image, l represents the distance of z and z' , ϵ represents a predetermined threshold that controls the similarity between z and z' .

In the identity hiding module, we hope to hide the identity of the target face in $X_{F_{adv}}$, so that the user's final published image is visually and identically consistent with X_F (Visual perception of identity is consistent with the source face). This operation makes it difficult for attackers to obtain the identity of the target face, thus achieving protection of the target identity. To achieve this goal, we introduce a user key to manipulate the latent variable of $X_{F_{adv}}$. We control the optimization goal of operator $\mathcal{O}$ through the following two losses.

Identity Consistency Loss: We introduce an identity consistency loss to make the reconstructed facial image's identity consistent with the source face. For simplicity, we set:

$$\mathcal{L}_{id} = \cos(R(X_{F_k}), R(X_s), 1), \quad (11)$$

where $X_{F_k} = \mathcal{G}_2(\mathcal{O}(\mathcal{E}(X_{F_{adv}}), k))$, R is the face recognition model (like CosFace), $\mathcal{E}$ is the pre-trained encoder, $\mathcal{G}_2$ is the pre-trained StyleGAN inversion generator and $\mathcal{O}$ is the latent space operator.

Reconstruction Loss: Since our goal is to obtain reconstructed image X_{F_k} that is visually close to adversarial swapped facial image $X_{F_{adv}}$. To maintain visual quality, adversarial images should be as similar as possible to the inputting $X_{F_{adv}}$ and we use reconstruction loss to maintain visual quality.

$$\mathcal{L}_{rec} = \frac{1}{2} \|X_{F_k} - X_{F_{adv}}\|_2^2. \quad (12)$$

3) Identity Authentication Module: In the identity authentication stage, driven by the correct key, the reconstructed face maintains its original appearance but has the identity of the target face. When using the wrong key for identity verification, the reconstructed facial image will be distorted and the identity of the target face cannot be traced.

Traceability Loss: If the correct key k_1 is utilized for verification, the reconstructed face belongs to the same identity as the target face. Nevertheless, in the event that an incorrect key k_2 is employed, the original face's identity cannot be reinstated. To achieve this goal, the traceability loss is introduced:

$$\begin{aligned} \mathcal{L}_{tra} = & \cos(R(\mathcal{G}_2(\mathcal{O}(\mathcal{E}(X_{reF_{k_1}}), k_1))), R(X_t), 1) \\ & + \cos(R(\mathcal{G}_2(\mathcal{O}(\mathcal{E}(X_{reF_{k_1}}), k_2))), R(X_t), -1), \end{aligned} \quad (13)$$

where $k_1 \neq k_2$, $X_{reF_{k_1}}$ is generated by k_1 .

Appearance Loss: To trace the source identity without revealing the appearance of the source, when using the correct key for authentication, the reconstructed image appearance is consistent with X_F . When the adversary attempts to authenticate with the wrong key, the appearance of the reconstructed image changes. To achieve this goal, we introduced appearance loss:

$$\mathcal{L}_{appe} = 1/2 \|\mathcal{G}_2(\mathcal{O}(\mathcal{E}(X_{reF_{k_1}}), k_1)) - X_F\|_2^2. \quad (14)$$

Total Loss: Our goal at this stage is a weighted average of the above loss functions, as shown below:

$$\mathcal{L}_{total2} = \mathcal{L}_{id} + \mathcal{L}_{rec} + \mathcal{L}_{tra} + \mathcal{L}_{appe}. \quad (15)$$

V. EXPERIMENTS

In this section, we provide a detailed introduction to our experimental settings, evaluate the performance of the proposed

method through extensive experiments and validate the safety of the method through security analysis.

A. Implementation Details

We provide a detailed introduction to the implementation details of our method and the deployment of experiments.

Datasets: Our model is trained and evaluated on LFW [33]. LFW has 13233 face images belonging to 5749 identities. We perform an 8:2 split according to the training and testing identities. Two images with different identities are divided into a group: one is designated as the source image (used to provide identity features for migration), and the other is designated as the target image (used to provide non-identity attributes such as pose and background), respectively.

CelebA is a publicly available large-scale celebrity face dataset, consisting of 202599 images across 10177 distinct identities. It covers diverse poses and expressions, enabling it to provide abundant test samples.

Pre-trained Models: To evaluate the generalization of the proposed method, we employ SimSwap [34], FaceShifter [4] and FSGAN [1] for face-swapping in our experiment and use pre-trained CosFace [35], ArcFace [36], and SphereFace [37] as authentication models. We use the closed box method for experiments, which means we do not need to know the internal structure, parameters and gradients of the face-swapping models and face recognition models.

Experimental Settings: We first train the identity transfer module, and then apply it to subsequent modules to complete the tasks of identity hiding and authentication. In the identity transfer module, the model is trained on the Adam optimizer for 50 epochs, where $\beta_1 = 0.5$, $\beta_2 = 0.999$ and set the learning rate as 1×10^{-4} . Based on experience, the tradeoff parameters of $\mathcal{L}_{total1}$ are set to $\lambda_{per} = 1$, $\lambda_{id} = 10$ and $\sigma = 5$.

In the identity hiding and authentication stages, we use a pre-trained latent encoder [16] and pre-trained StyleGAN's inversion model [38] as encoder $\mathcal{E}$ and generator $\mathcal{G}$, respectively. We use the Adam optimizer to train our network with $\beta_1 = 0.5$, $\beta_2 = 0.999$ and set the learning rate to 1×10^{-4} . The range of ϵ is set to $[0.01, 0.05]$. We define k as an 8-bit password. All of the experiments are conducted on RTX3090 GPU.

Evaluation Metrics: Multiple metrics were selected to evaluate the performance of the IPST model. To evaluate the quality of the synthesized swapped facial images, peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) are selected as evaluation metrics. Learned perceptual image patch similarity (LPIPS) distance is applied to measure visual similarity which is closer to human perception than traditional metrics.

State of the Arts: To our knowledge, there is currently no research on authorized face-swapping authentication in the countermeasures against Deepfake. Therefore, we cannot directly compare the identification of source IDs with any existing methods. Despite all this, we still compare our method with methods of active tracking, including SepMark [19], SIDT [21] and Anti-Forgery [17]. Wu et al. [19] embed robust and fragile watermarks in forged images, and uses the integrity of the watermarks for forgery detection and traceability. Lin et al. [21]

TABLE II
COMPARISON OF SOURCE PRESERVING FUNCTIONS

Methods	Beforehand	Afterward	Face-swapping Available	Traceability	Detection	Key-Controlled
SepMark	✓	✗	✓	✓	✓	✗
SIDT	✗	✓	✓	✓	✗	✗
Anti-Forgery	✓	✓	✗	✗	✗	✗
Ours	✗	✓	✓	✓	✓	✓

TABLE III
IMAGE QUALITY EVALUATION RESULTS

FS	Image pair	SSIM ↑	PSNR ↑	LPIPS↓
SimSwap	$\langle X_F, X_{F_{adv}} \rangle$	0.9865	38.47	0.022
	$\langle X_F, X_{F_k} \rangle$	0.9812	36.23	0.028
	$\langle X_F, X_{reF_k} \rangle$	0.9657	39.14	0.019
FaceShifter	$\langle X_F, X_{F_{adv}} \rangle$	0.9738	35.43	0.029
	$\langle X_F, X_{F_k} \rangle$	0.9719	36.94	0.035
	$\langle X_F, X_{reF_k} \rangle$	0.9527	35.10	0.029
FSGAN	$\langle X_F, X_{F_{adv}} \rangle$	0.9625	38.47	0.035
	$\langle X_F, X_{F_k} \rangle$	0.9534	36.23	0.038
	$\langle X_F, X_{reF_k} \rangle$	0.9422	39.14	0.041

embed the source facial image into a forged image and the source facial image can be restored through a decoder. Although the modified method can authenticate the source identity, it will expose the source surface during the recognition process, making it unable to protect at the sensory level. Wang et al. [17] add adversarial noise to the original image, causing severe visual distortion in the swapped facial images. This operation prevents the occurrence of face-swapping at the source. The comparison results are shown in Table II.

B. Experimental Results

We use pre-trained face recognition models to evaluate the performance of identity conversion. To verify the generality of the proposed method, different face-swapping networks and face recognition networks are chosen during the evaluation process. To verify the superiority of the proposed method, it is compared with advanced adversarial attack methods and SOTA in terms of visual quality and identity verification.

1) *Image Quality Assessment*: To evaluate the impact of IPST operations in identity preservation and tracking on the quality of the swapped facial images, we evaluate the performance of our method under different face-swapping models. Evaluate visual loss by separately calculating SSIM, PSNR, MSE and LPIPS between the following image pairs: X_F (identity from X_s , appearance from X_t) and $X_{F_{adv}}$, X_F , X_{F_k} , X_F and X_{reF_k} . The image quality evaluation results are shown in Table III.

From Table III, our method has reconstructed visual quality for various face-swapping models. We will evaluate the visual quality of generating swapped facial images using the proposed method and SIDT [21] (considering that it is also an afterward). In this experiment, considering the differences in methods, we only evaluated the image quality between the original swapped facial image generated by the FaceShifter [4] network and the

TABLE IV
COMPARISON OF IMAGE QUALITY WITH SIDT

Metrics	SSIM ↑	PSNR ↑	LPIPS↓
SIDT	0.9651	33.01	0.038
Ours	0.9719	36.94	0.035

TABLE V
EVALUATION RESULTS OF IDENTITY-PRESERVING

FS Method	Recognition Model	LFW		CelebA	
		$\langle X_{F_{adv}}, X_t \rangle$	$\langle X_{F_k}, X_s \rangle$	$\langle X_{F_{adv}}, X_t \rangle$	$\langle X_{F_k}, X_s \rangle$
SimSwap	CosFace	0.832	0.839	0.812	0.822
	ArcFace	0.859	0.829	0.827	0.789
	SphereFace	0.793	0.781	0.761	0.802
FaceShifter	CosFace	0.788	0.783	0.821	0.786
	ArcFace	0.793	0.825	0.825	0.808
	SphereFace	0.830	0.783	0.796	0.771
FSGAN	CosFace	0.810	0.799	0.843	0.802
	ArcFace	0.836	0.811	0.806	0.781
	SphereFace	0.789	0.804	0.822	0.817

protected swapped facial image. Table IV reports the visual quality evaluation on identity-preserving of the source. It can be seen that compared to SIDT, the swapped facial image protected by our method have better evaluations in SSIM, PSNR and LPIPS.

2) *Identity-Preserving Assessment*: Our identity-preserving process is implemented by two modules: identity transfer and identity hiding. We add weak adversarial perturbations to the swapped facial image to make $X_{F_{adv}}$ consistent with the target face in terms of identity. To prevent unauthorized access, we modify the latent variables in the latent space using the user key, so that the identity of the reconstructed image X_{F_k} is consistent with the source face. Three different face-swapping models are selected for the evaluation of identity protection. We evaluated three pre-trained facial recognition models on the LFW and CelebA datasets, and Table V reports the changes in identity during both identity transfer and identity hiding processes. The collaborative use of two stages ensures that the protected swapped facial image still matches the source face in terms of identity, which can prevent adversaries from maliciously accessing the target identity. The experimental results indicate that the target identity can be protected under different face-swapping models.

The experimental results show that IPST can protect and hide the target identity under different face-swapping models, and has good performance across datasets.

3) *Identity Authentication Assessment*: The goal of the proposed method is to determine the ownership of identity by

TABLE VI
EVALUATION RESULTS OF IDENTITY TRACEABILITY

FS Method	Recognition Model	LFW		CelebA	
		$\langle X_{reF_{k_1}}, X_t \rangle$	$\langle X_{reF_{k_2}}, X_t \rangle$	$\langle X_{reF_{k_1}}, X_t \rangle$	$\langle X_{reF_{k_2}}, X_t \rangle$
SimSwap	CosFace	0.818	0.483	0.852	0.451
	ArcFace	0.834	0.628	0.876	0.592
	SphereFace	0.791	0.555	0.823	0.527
FaceShifter	CosFace	0.789	0.382	0.831	0.465
	ArcFace	0.791	0.538	0.845	0.512
	SphereFace	0.809	0.427	0.837	0.401
FSGAN	CosFace	0.821	0.519	0.848	0.493
	ArcFace	0.882	0.601	0.795	0.578
	SphereFace	0.799	0.342	0.829	0.426

tracking when a face-swapping dispute occurs. When the correct key is used for authentication, the reconstructed face matches the identity of the target face. When an attacker attempts to track the target identity with the wrong key, authentication will fail, meaning it will not be able to match the identity of the target face. To evaluate the traceability ability of our method, we evaluate the recognition performance of the face recovered with the correct key $X_{reF_{k_1}}$ and the face recovered with the wrong key $X_{reF_{k_2}}$ against the original face X_s on the LFW dataset. Three face recognition model are used to match the aforementioned facial pairs. The experimental results are shown in Table VI. It can be seen that IPST has achieved satisfactory average face recognition rates on all three face recognition devices under the validation of LFW and CelebA datasets, and is effective for different face-swapping models. This indicates the generality of the proposed IPST in different facial recognition scenarios and its effectiveness in different face-swapping models.

4) *Comparisons With Adversarial Perturbations:* The identity preserving process is designed to transform the identity of the swapped facial image from the source face to the target face, and then back to the source, without altering its original appearance. We achieve this through a simulated adversarial attack and a latent space editing. Without considering key control, the above identity conversion process can be achieved through two simulated adversarial attacks. To verify the superiority of our method, we implement the identity transfer process through two pixel-level adversarial attacks and compare it with our method in terms of recognition rate and image quality. The famous adversarial methods PGD [39], TIP-IM [40], and AdvFace [41] are compared with the proposed methods and evaluate using three commonly used facial recognition models. We compare the visual quality and recognition accuracy between the original swapped facial image generated by SimSwap and the adversarial image and the experimental results are shown in Table X. The experimental results show that our method performs better in image quality and identity conversion performance. The comparative methods simulate identity transfer and identity hiding processes through two pixel level perturbations, which not only sacrifices security performance but also leads to significant sacrifices in recognition accuracy and image quality. However, the proposed method ensures the maintenance of image quality and enhances security.

5) *Generalization Analysis:* To demonstrate the universality of the proposed method and its ability to adapt to different

face-swapping and recognition scenarios, we conduct extensive experiments to verify the performance of our method. Three different face-swapping models and face recognition models were selected to evaluate the effectiveness and universality of the proposed model. The experimental results show that the proposed method can protect swapped facial images under different face-swapping models and can be applied to different face recognition interfaces. The above analysis can be confirmed by the experimental results in Tables III, V, and VI.

6) *Robustness Evaluation:* We select five common post-processing operations (JPEG compression, Gaussian noise, Gaussian blur, cropping, scaling) and set different intensities for each. Table IX presents the traceability performance under varying post-processing intensities.

Experimental results show that in mild post-processing of core applicable scenarios, the IPST maintains a traceability success rate of ≥ 0.815 , with ≥ 0.846 in typical scenarios such as JPEG 90% compression, 5% edge cropping, and 3×3 Gaussian blur. Post-processing in these scenarios does not damage image integrity, and the IPST's robustness is sufficient to support traceability. For moderate and extreme post-processing scenarios, the IPST's traceability success rate drops to 0.725, and even below 0.6 in some extreme cases. It is emphasized that swapped facial images in such scenarios suffer severe quality loss (e.g., block distortion caused by JPEG 0.5 compression, facial feature loss due to 10% center cropping), resulting in a significant loss of visual usability and practical application value, which is not the objective of IPST.

C. Security Analysis

We conduct security analysis by discussing the impact of different key lengths on performance, key fault tolerance and attackers attempting to authenticate with incorrect keys. In addition, we verify the necessity of the key by simulating attacks on proxy networks.

1) *Impact of Key-Length:* To discuss the impact of different key lengths on performance, we set the keys to 8-bit, 32-bit, and 128-bit for identity protection and identity verification of swapped facial images generated by SimSwap, and evaluate recognition performance using CosFace. Table XI reports the impact of keys of different lengths on the protection process and authentication process. From Table XI, it is shown that different key lengths have a slight impact on the quality and performance of reconstructed images. This indicates that our method can adapt to different key lengths and we can improve security performance by increasing the key length in practical applications.

2) *Error Rate of Key:* To demonstrate the security of the proposed key strategy, the fault tolerance of keys of different lengths is tested. We evaluate the number of bits that can accept incorrect keys for 8-bit, 32-bit and 128-bit keys, respectively. As shown in Table VII, When the number of erroneous bits is 0, the Structure Similarity Index Measure (SSIM) values of the images all exceed 0.90, the Learned Perceptual Image Patch Similarity (LPIPS) values are all below 0.1, the Peak Signal-to-Noise Ratio (PSNR) values exceed 30 dB, and the identity distances (Id-distance)

TABLE VII
ERROR RATE OF DIFFERENT LENGTH KEYS

key length	Error bit	PSNR $\uparrow$	LPIPS $\downarrow$	SSIM $\uparrow$	Id-sim $\uparrow$
8-bit	0	31.14	0.09	0.91	0.839
	1	20.85	0.22	0.62	0.600
32-bit	0	32.14	0.01	0.93	0.872
	5	23.85	0.22	0.65	0.582
	10	20.88	0.24	0.61	0.612
128-bit	0	30.88	0.09	0.90	0.822
	5	21.83	0.18	0.71	0.631
	10	19.94	0.19	0.64	0.472

calculated using features extracted by CosFace are all higher than 0.82. This indicates that when traceability is performed with the correct key, the visual quality of the images remains stable, and the target identity can be effectively recovered. When the number of erroneous bits is ≥ 1 , the SSIM values drop sharply to below 0.65, the LPIPS values are significantly higher than 0.1, the PSNR values are below 24 dB, and the Id-distance values are lower than 0.63. This demonstrates that the recovered face images exhibit obvious visual distortion, and the identity recovery task fails. The proposed key strategy has a low fault tolerance, indicating that our method is secure and attackers find it difficult to bypass the key or attempt to obtain the identity of the target face using the wrong key.

3) *Security Assessment*: To protect the identity of the target face from being easily accessed, we propose a protection and traceability method controlled by the key. In the event of a face-swapping dispute, users can use the correct key to restore the identity of the target face without compromising the appearance of the swapped facial image. However, adversaries may attempt to illegally obtain the identity of the target face using the wrong key, thereby using some methods to compromise the performance of authentication. To evaluate the security of the proposed method, a user key is used to protect the swapped facial image, and then simulate an attacker's attack using a randomly generated incorrect key. The experimental results are shown in Tables VI and XI, indicating that when attackers use the wrong key for identity verification, they are unable to obtain the identity of the target face and the reconstructed image will suffer significant visual loss. Fig. 4 shows some examples, from left to right, which are the target face, target face, identity-preserving image, image reconstructed with the correct key and image reconstructed with the wrong key. When the key is incorrect, the appearance of the reconstructed image will undergo significant changes or cause serious visual loss.

4) *Necessity of Key*: Unlike schemes relying solely on the secrecy of encoder/recovery network hyperparameters, the key in IPST functions as a dynamically bindable credential at the application layer, undertaking two irreplaceable roles: identity hiding and access control. A defining feature of this key is its non-uniqueness, legitimate users are granted the freedom to select or replace keys at will, and the binding relationship between identity and key is not fixed by the model itself but determined entirely by user configurations. Consequently, the same target identity can be protected by different keys, resulting in distinct

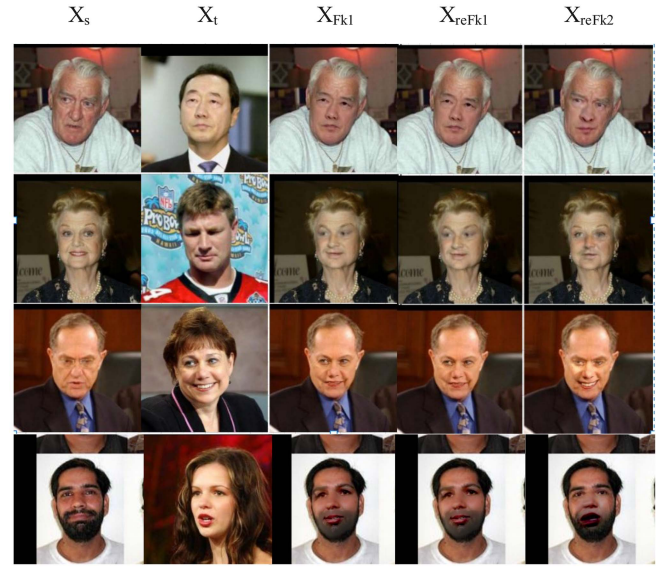


Fig. 4. Exemplary results for the recovery with correct or wrong password: In each row, the first image is the original image, the second image is the target image, the third is the output forgery image, the fourth is the image of correct key authentication and the fifth is the image of incorrect key authentication.

TABLE VIII
IMPACT OF DIFFERENT ϵ ON PERFORMANCE

ϵ	SSIM	CosFace	ArcFace
0.01	0.922	0.725	0.698
0.02	0.917	0.779	0.798
0.03	0.895	0.832	0.859
0.04	0.862	0.842	0.863
0.05	0.833	0.851	0.861

protected identity representations. For adversaries, mere knowledge of the model's hyperparameters (e.g., network structure, training datasets) is insufficient to infer the dynamic binding rule of a specific key. Without possessing the corresponding key, they are unable to recover the hidden target identity.

We randomly select 11000 different identities and generate IPST protected swapped facial images using the same key. We select 10000 protected identities and swapped facial images as leaked data for adversary to train a proxy network that is consistent with the IPST traceability function (directly extracting protected identities from protected images). The remaining 1000 protected images are used as Test Set A to verify the attack performance under the initial key. The user updates the keys of the original 1000 test identities and generates new protected images as Test Set B. The proxy network attack performance is tested again, with evaluation metrics including average identity similarity and attack success rate (similarity ≥ 0.75).

As shown in Table XIII, the surrogate network can only trace identities protected by the key of the leaked data, and its attack success rate drops to 9.5% for the updated Test Set B. Key updates do not affect legitimate users' normal tracing through IPST, but render the surrogate network trained on old data completely ineffective, indicating that the dynamic key mechanism can block proxy network attacks.

TABLE IX
TRACEABILITY SUCCESS RATE (TSR) OF IPST UNDER COMMON IMAGE TRANSFORMATIONS

Image Transformation	Intensity Level	TSR (Mean±SD)	Scenario Adaptability Analysis
JPEG Compression	Quality=90 (Mild)	0.863±0.014	Video post-production, stable traceability
	Quality=70 (Medium)	0.725±0.022	Social platform compression, performance degradation
	Quality=50 (Extreme)	0.587±0.028	Extreme compression, no practical usability
Gaussian Noise	$\sigma=0.05$ (Mild)	0.832±0.016	Minor interference, meets scenario requirements
	$\sigma=0.10$ (Medium)	0.654±0.025	Obvious noise, not the target scenario
Gaussian Blur	Kernel=3×3 (Mild)	0.815±0.018	Mild optimization, effective traceability
	Kernel=5×5 (Medium)	0.593±0.031	Severe blur, no practical
Cropping	5% (Edge, Mild)	0.846±0.015	Minor edge cropping, stable performance
	10% (Center, Extreme)	0.612±0.029	Core area cropping, lost usability
Scaling	0.8× (Mild Downscaling)	0.827±0.017	Mild resizing, good traceability
	0.5× (Extreme Downscaling)	0.638±0.026	Extreme resizing, no practical use

TABLE X
RESULTS OF PERFORMANCE COMPARISON WITH OTHER ADVERSARIES

Methods	Image Quality Evaluation			Authentication Assessment		
	SSIM ↑	PSNR ↑	LPIPS ↓	CosFace	ArcFace	ShereFace
PGD	0.535	20.27	0.42	0.368	0.353	0.102
TIP-IM	0.838	28.39	0.16	0.544	0.402	0.375
AdvFace	0.871	28.80	0.13	0.724	0.622	0.755
IPST	0.895	31.14	0.09	0.832	0.859	0.793

TABLE XI
IMPACT OF DIFFERENT KEY LENGTHS

Methods	Image Quality Evaluation			Authentication Assessment		
	SSIM ↑	PSNR ↑	LPIPS ↓	CosFace	ArcFace	ShereFace
8-bit	0.895	31.14	0.09	0.832	0.859	0.793
32-bit	0.872	32.01	0.15	0.811	0.833	0.801
128-bit	0.887	30.88	0.09	0.826	0.824	0.789

TABLE XII
ABLATION STUDIES FOR IPST

Methods	Image Quality Evaluation			Authentication Assessment		
	SSIM ↑	PSNR ↑	LPIPS ↓	CosFace	ArcFace	ShereFace
w/o $\mathcal{L}_{id}$	0.695	24.29	0.22	0.398	0.412	0.329
w/o $\mathcal{L}_{rec}$	0.538	25.39	0.28	0.544	0.402	0.408
IPST	0.895	31.14	0.09	0.832	0.859	0.793

TABLE XIII
EXPERIMENTAL RESULTS OF SURROGATE NETWORK ATTACK

Test Scenario	Id-sim	Attack Success Rate
IPST (Test Set A)	0.88	92.3%
IPST (Test Set B)	0.89	93.1%
Surrogate Network (Test Set A)	0.79	67.5%
Surrogate Network (Test Set B)	0.62	9.5%

TABLE XIV
COMPUTATIONAL COST COMPARISON AMONG IPST, SEPMark, AND SIDT

Technique Type	Core Operation	Processing Time (s)
SepMark	Deep Separable Watermark Embedding	0.38 ± 0.03
SIDT	Original Face Embedding and Visual Reconstruction	4.7 ± 0.6
IPST	Identity Transfer and Latent Space Editing	2.3 ± 0.4

D. Efficiency Analysis

To fully verify the efficiency of IPST, Table XIV further includes SIDT [21], a visual reconstruction-based traceability scheme. It conducts quantitative comparisons with the robust

watermarking technique SepMark [19] and IPST on a unified hardware platform. The test data covers 1000 face-swapping samples from the LFW and CelebA datasets.

As shown in the table, the robust watermarking technique SepMark [19] takes only 0.38 seconds per image. It is suitable for ultra-large-scale scenarios with low security requirements such as social media content sharing. SIDT's core operation has a per-image processing time of 4.7 seconds. Its efficiency is significantly lower than other schemes. IPST's per-image processing time is 2.3 seconds, higher than that of robust watermarking but lower than SIDT. IPST is designed for small-to-medium-scale authorized face-swapping scenarios such as film digital doubles and medical case anonymization. Its efficiency meets the needs of target scenarios. The feature of one-time processing supporting unlimited traceability further reduces long-term comprehensive costs. More importantly, IPST does not need to reconstruct the visual information of the original face. It ensures the validity of traceability evidence and avoids the exposure of the original face's appearance. At the same time, it supports dynamic key updates. If a key is suspected of being leaked, it can be replaced directly without retraining the model. Once the code of watermarking technology is leaked, it becomes permanently incorrect. Alternatively, it may introduce the training cost of fine-tuning the model.

E. Ablation Study

We conduct ablation studies on the values of each loss and ϵ during the preserving phase.

The Effectiveness of Each Loss: To discuss the impact of image quality loss and identity consistency loss, we remove each loss from the overall training objective and quantitative evaluation results, respectively (Image quality evaluation X_t and X_{Fre} and the accuracy of identity recognition is between X_s and X_{Fre}) are shown in Table XII. w/o $\mathcal{L}_{id}$ shows that $\mathcal{L}_{id}$ is necessary to achieve identity transfer and affect the quality of reconstructed images. w/o $\mathcal{L}_{rec}$ shows that without reconstruction loss $\mathcal{L}_{rec}$ constraints, the similarity between the obtained image and the target image is low and it also affects the accuracy of the target identity recognition results.

Selection of ϵ : To maintain the similarity in appearance between the reconstructed image and the input image, the hyperparameter ϵ is used to constrain the range of modifications to the latent variables controlled by the key. From Table VIII, when the value of ϵ increases, the change rate of latent variables also grows and the identity distance between the reconstructed image and the target image decreases. However, the increase in recognition performance comes at the cost of sacrificing structural similarity. To balance the contradiction between the structural similarity of reconstructed images and recognition accuracy, ϵ is set to 0.03.

VI. CONCLUSION

In this paper, we proposed a key-driven Identity Protection and Secure Traceability (IPST) framework for face swapping under privacy-preserving scenarios. Different from conventional approaches that rely on external data embedding, the proposed method establishes an implicit association between generated images and target identities in the facial feature space, enabling identity-level binding while preserving visual anonymity. IPST employs a three-stage mechanism encompassing identity injection, identity concealment, and authorization recovery, binding the target identity to the face-swapped image in the feature space and ensuring that identity verification only occurs under authorized conditions. Extensive experimental results demonstrate that IPST achieves a good balance between image visual quality, identity binding effectiveness, and resistance to unauthorized recovery. Furthermore, the proposed pluggable design can be integrated into existing face-swapping systems, thus supporting practical deployment in privacy-sensitive scenarios. The proposed method relies on pre-trained face recognition models for identity feature extraction, and its robustness remains limited under strong post-processing conditions. In future work, we will introduce a multi-scale adaptive feature extraction mechanism to dynamically optimize feature representation, improving robustness under complex real-world transformations.

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